Math 110 — Week 9

Trace and inner products (Notes Ch. 16–17, LADR 8D, 6A, HW9)

§1 Facts used from earlier weeks

V is finite-dimensional over $\mathbb{F}$, and $n = \dim V$.

(T1) **Change of basis.** If $A = \mathcal{M}_{\mathcal{B}}(T)$ and $A' = \mathcal{M}_{\mathcal{C}}(T)$ are the matrices of $T \in \mathcal{L}(V)$ in two bases, then $A' = P^{-1}AP$, where column j of the invertible matrix P holds the $\mathcal{B}$-coordinates of the j-th vector of $\mathcal{C}$.

Why. Both sides send the $\mathcal{C}$-coordinates of v to the $\mathcal{C}$-coordinates of Tv: P converts $\mathcal{C}$-coordinates to $\mathcal{B}$-coordinates, A applies T, and P^{-1} converts back.

(T2) **Jordan form.** Over $\mathbb{C}$, every $T \in \mathcal{L}(V)$ has a basis in which its matrix is upper-triangular, with the eigenvalues on the diagonal, each repeated as often as its algebraic multiplicity $\dim G(\lambda, T)$ (Week 8).

(T3) **Characteristic polynomial.** Over $\mathbb{C}$, $\chi_T(z) = \prod_{j=1}^{n}(z - \lambda_j)$, where $\lambda_1, \dots, \lambda_n$ are the eigenvalues listed with algebraic multiplicity (Week 8).

(T4) **Complex numbers.** $\overline{z+w} = \bar{z} + \bar{w}$, $\overline{zw} = \bar{z}\,\bar{w}$, $z\bar{z} = |z|^2$, $z + \bar{z} = 2\operatorname{Re} z$, and $\operatorname{Re} z \le |z|$. If a polynomial has real coefficients, its non-real roots come in conjugate pairs $\lambda, \bar{\lambda}$ with the same multiplicity.

Why. The first four are direct computations with $z = a + bi$. $\operatorname{Re} z = a \le \sqrt{a^2 + b^2} = |z|$. If p has real coefficients, then $\overline{p(\lambda)} = p(\bar{\lambda})$, so $p(\lambda) = 0 \iff p(\bar{\lambda}) = 0$; applying this to the quotient $p/(x - \lambda)(x - \bar{\lambda})$, which again has real coefficients, gives equal multiplicities.

(T5) **A nonnegative continuous function with integral 0 is 0.** If $g\colon [a, b] \to \mathbb{R}$ is continuous, $g \ge 0$ and $\int_a^b g = 0$, then $g = 0$.

Why. If $g(c) > 0$, continuity gives an interval around c on which $g > g(c)/2$, and the integral over that interval alone is positive.

§2 This week

§2.1 The trace of a matrix (Ch. 16)

The question. Jordan form showed that a good basis reveals a lot about an operator. Is there a number we can read off *any* matrix of T that comes out the same in every basis?

Definition (Def. 41 / LADR 8.47)**.** The **trace** of an $n \times n$ matrix $A = (a_{ij})$ is the sum of its diagonal entries: $\operatorname{tr} A = a_{11} + \cdots + a_{nn}$.

For example,

$$\operatorname{tr}\begin{pmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{pmatrix} = 1+5+9 = 15.$$

Trace is linear: $\operatorname{tr}(A+B) = \operatorname{tr}A + \operatorname{tr}B$ and $\operatorname{tr}(cA) = c\operatorname{tr}A$, because the diagonal of $A+B$ is the sum of the diagonals. It is not multiplicative: $\operatorname{tr}(AB)$ is usually not $\operatorname{tr}A \cdot \operatorname{tr}B$ (take $A = B = I_2$: $\operatorname{tr}(AB) = 2$, $\operatorname{tr}A \cdot \operatorname{tr}B = 4$). What it does satisfy is the following.

Theorem (Prop. 78 / LADR 8.49: the trace of a product does not depend on the order)
If A is $m \times n$ and B is $n \times m$, then $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

The idea

Write out both traces as sums of products of entries. The diagonal entry $(AB)_{ii}$ is row i of A times column i of B, which is $\sum_j a_{ij}b_{ji}$. So $\operatorname{tr}(AB)$ is the sum of all the products $a_{ij}b_{ji}$. The same products appear in $\operatorname{tr}(BA)$, only grouped by j instead of by i. Nothing else is needed: the two traces are one finite sum added in two different orders.

Proof. 1. By the definition of matrix multiplication, $(AB)_{ii} = \sum_{j=1}^{n} a_{ij}b_{ji}$ for $i = 1, \dots, m$, so

$$\operatorname{tr}(AB) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}b_{ji}.$$

2. In the same way $(BA)_{jj} = \sum_{i=1}^{m} b_{ji}a_{ij}$ for $j = 1, \dots, n$, so

$$\operatorname{tr}(BA) = \sum_{j=1}^{n}\sum_{i=1}^{m} b_{ji}a_{ij}.$$

3. Both are the sum of the mn numbers $a_{ij}b_{ji}$, over all pairs (i, j). A finite sum does not depend on the order of its terms, so the two traces are equal.

□

	$j=1$	$j=2$	$j=3$	add along the row
$i=1$	$a_{11}b_{11}$	$a_{12}b_{21}$	$a_{13}b_{31}$	$(AB)_{11}$
$i=2$	$a_{21}b_{12}$	$a_{22}b_{22}$	$a_{23}b_{32}$	$(AB)_{22}$
add down the column	$(BA)_{11}$	$(BA)_{22}$	$(BA)_{33}$	total of all six

The case $m = 2$, $n = 3$. The six products $a_{ij}b_{ji}$, one for each pair (i, j). Adding along the rows gives the diagonal entries of AB; adding down the columns gives the diagonal entries of BA. Both totals are the sum of all six products.

Example: for

$$A = \begin{pmatrix}1 & 2 & 0\\ 0 & 1 & 3\end{pmatrix}, \quad B = \begin{pmatrix}1 & 0\\ 2 & 1\\ 0 & 1\end{pmatrix}: \qquad AB = \begin{pmatrix}5 & 2\\ 2 & 4\end{pmatrix}, \quad BA = \begin{pmatrix}1 & 2 & 0\\ 2 & 5 & 3\\ 0 & 1 & 3\end{pmatrix},$$

and both traces are 9, although AB is 2×2 and BA is 3×3.

Similar matrices have the same trace. If P is invertible, apply Prop. 78 to the two matrices P^{-1} and AP:

$$\operatorname{tr}(P^{-1}AP) = \operatorname{tr}\big(P^{-1}(AP)\big) = \operatorname{tr}\big((AP)P^{-1}\big) = \operatorname{tr}A.$$

Cyclic order only. Applying Prop. 78 to A and BC gives $\operatorname{tr}(ABC) = \operatorname{tr}(BCA)$, and once more $= \operatorname{tr}(CAB)$: moving the first factor to the end is allowed. Swapping two factors is not. With the 2×2 matrix units E_{11}, E_{12}, E_{21} (E_{ij} has a single 1 in position (i, j)):

$$\operatorname{tr}(E_{11}E_{12}E_{21}) = \operatorname{tr}E_{11} = 1, \qquad \operatorname{tr}(E_{11}E_{21}E_{12}) = \operatorname{tr}(0 \cdot E_{12}) = 0.$$

§2.2 The trace of an operator (Ch. 16)

Definition (Def. 42 / LADR 8.50, 8.51)**.** For $T \in \mathcal{L}(V)$, choose any basis $\mathcal{B}$ of V and define $\operatorname{tr} T = \operatorname{tr} \mathcal{M}_{\mathcal{B}}(T)$.

This does not depend on the basis: by (T1) the matrices of T in two bases are similar, and similar matrices have the same trace. So $\operatorname{tr} T$ is computed from coordinates but is a property of T alone. In particular $\operatorname{tr} I_V = n$, and tr is linear on $\mathcal{L}(V)$.

Theorem (Prop. 79 / LADR 8.52: over $\mathbb{C}$, trace is the sum of the eigenvalues)
If V is a complex vector space and $T \in \mathcal{L}(V)$, then $\operatorname{tr} T$ is the sum of the eigenvalues of T, each counted with its algebraic multiplicity. If the distinct eigenvalues are $\lambda_1, \dots, \lambda_r$ with algebraic multiplicities $m_1, \dots, m_r$, then $\operatorname{tr} T = m_1\lambda_1 + \cdots + m_r\lambda_r$.

The idea

The trace may be computed in any basis, so compute it in the basis where the eigenvalues are visible: the Jordan basis, whose diagonal is exactly the list of eigenvalues with multiplicities.

Proof. 1. By (T2) there is a basis in which $\mathcal{M}(T)$ is upper-triangular with diagonal entries $\lambda_1, \dots, \lambda_1, \dots, \lambda_r, \dots, \lambda_r$, where λ_k appears m_k times.

2. $\operatorname{tr} T$ is the trace of this matrix, which is the sum of its diagonal entries, $m_1\lambda_1 + \cdots + m_r\lambda_r$.

□

- Ex. 64: the eigenvalues of
$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix}$$
are roots of the cubic $\lambda^3 - 12\lambda^2 + 15\lambda - 2$, which are unpleasant to write down. Their sum is still immediate:
$$\lambda_1 + \lambda_2 + \lambda_3 = \operatorname{tr} A = 1 + 2 + 9 = 12.$$
- **Trace in the characteristic polynomial.** Expanding $\chi_T(z) = \prod_{j=1}^{n}(z - \lambda_j)$ (T3), the terms with z^{n-1} come from choosing z in all factors but one and $-\lambda_j$ in the remaining one. So the coefficient of z^{n-1} is $-(\lambda_1 + \cdots + \lambda_n) = -\operatorname{tr} T$. In the cubic of Ex. 64 this coefficient is -12.
- **Real matrices.** A real matrix A is also an operator on $\mathbb{C}^n$, and its trace is the same number. Its characteristic polynomial has real coefficients, so its non-real eigenvalues come in pairs $\lambda, \bar{\lambda}$ (T4), and each pair contributes $\lambda + \bar{\lambda} = 2\operatorname{Re}\lambda$ to the trace. Example: rotation of $\mathbb{R}^2$ by θ has eigenvalues $e^{\pm i\theta}$ and trace $2\cos\theta$, which is the sum of its diagonal entries $\cos\theta + \cos\theta$.

§2.3 Commutators have trace zero (Ch. 16)

Theorem (Prop. 80: commutators have trace 0)
For $S, T \in \mathcal{L}(V)$, $\operatorname{tr}(ST - TS) = 0$. Consequently, if $V \neq \{0\}$ and $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$, there are no $S, T \in \mathcal{L}(V)$ with $ST - TS = I$.

The idea

ST and TS are different operators, but Prop. 78 says they have the same trace, so their difference has trace 0. The identity has trace $n \neq 0$, so it cannot be a difference of this kind.

Proof. 1. Trace is linear, so $\operatorname{tr}(ST - TS) = \operatorname{tr}(ST) - \operatorname{tr}(TS)$. In any basis, $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$ and $\mathcal{M}(TS) = \mathcal{M}(T)\mathcal{M}(S)$, and these have the same trace by Prop. 78. So $\operatorname{tr}(ST - TS) = 0$.

2. $\operatorname{tr} I = n$, which is not 0 in $\mathbb{R}$ or $\mathbb{C}$ because $n \geq 1$. So $ST - TS = I$ would give $0 = n$, which is false.

□

Finite dimension is needed. On the infinite-dimensional space $\mathcal{P}(\mathbb{R})$, let D be differentiation and M multiplication by x. The product rule gives $D(xp) = p + xp'$, so $(DM - MD)p = p$ for every p: here $DM - MD = I$. (In quantum mechanics this is the relation between momentum and position.) There is no trace on $\mathcal{P}(\mathbb{R})$ to rule it out.

§2.4 Inner products (Ch. 17)

The question. So far the course has been algebraic: nothing has had a length or an angle. Lengths and angles in $\mathbb{R}^2$ come from the dot product. Which properties of the dot product are needed to do geometry in an arbitrary vector space, including a complex one?

From now on $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$. Over $\mathbb{R}$, ignore every complex conjugation below.

Definition (Def. 43 / LADR 6.2)**.** An **inner product** on V is a function $\langle \cdot, \cdot \rangle \colon V \times V \to \mathbb{F}$ such that for all $u, v, w \in V$ and $\lambda \in \mathbb{F}$:

(i) *positive definite:* $\langle v, v \rangle > 0$ for every $v \neq 0$;

(ii) *linear in the first slot:* $\langle \lambda u + v, w \rangle = \lambda \langle u, w \rangle + \langle v, w \rangle$;

(iii) *conjugate symmetric:* $\langle u, v \rangle = \overline{\langle v, u \rangle}$.

An **inner product space** is a vector space with a chosen inner product.

Why the conjugate. On $\mathbb{R}^n$ the dot product $\sum x_j y_j$ is positive on nonzero vectors, because $\sum x_j^2 > 0$. Copied to $\mathbb{C}^n$ without change, it fails: $(i, 1) \cdot (i, 1) = i^2 + 1^2 = 0$ although $(i, 1) \neq 0$. The fix is to conjugate the second vector, because $z\bar{z} = |z|^2 \geq 0$ (T4).

Consequences of the axioms.

- $\langle v, v \rangle$ is real, by (iii) with $u = v$.
- *Conjugate-linear in the second slot:* $\langle u, \lambda v + w \rangle = \bar{\lambda} \langle u, v \rangle + \langle u, w \rangle$. Proof: by (iii), (ii), and (iii) again with (T4),

$$\langle u, \lambda v + w \rangle = \overline{\langle \lambda v + w, u \rangle} = \overline{\lambda \langle v, u \rangle + \langle w, u \rangle} = \bar{\lambda}\, \overline{\langle v, u \rangle} + \overline{\langle w, u \rangle} = \bar{\lambda} \langle u, v \rangle + \langle u, w \rangle.$$

- $\langle 0, v \rangle = \langle v, 0 \rangle = 0$: by (ii), $\langle 0, v \rangle = \langle 0 \cdot 0, v \rangle = 0 \langle 0, v \rangle = 0$, and then (iii) gives the other.

Examples.

- Ex. 65, the **dot product** on $\mathbb{F}^n$: $\langle w, z \rangle = w_1 \bar{z}_1 + \cdots + w_n \bar{z}_n$. Positive definite because $\langle w, w \rangle = |w_1|^2 + \cdots + |w_n|^2$, which is 0 only when every $w_j = 0$. Properties (ii) and (iii) follow from (T4) term by term.
- Ex. 66, the L^2 **product** on continuous functions $[0, 1] \to \mathbb{C}$: $\langle f, g \rangle = \int_0^1 f(x) \overline{g(x)}\, dx$. Linearity and conjugate symmetry come from linearity of the integral. Positive definite: $\langle f, f \rangle = \int_0^1 |f|^2$, and if this is 0 then $|f|^2 = 0$ by (T5), so $f = 0$. The same works on any interval $[a, b]$; on an unbounded interval the integrals must be required to converge.
- Weights: on $\mathbb{R}^2$, $\langle x, y \rangle = x_1 y_1 + x_1 y_2 + x_2 y_1 + 2 x_2 y_2$ is an inner product. Linearity and symmetry are clear; for positivity complete the square:

$$\langle x, x \rangle = x_1^2 + 2 x_1 x_2 + 2 x_2^2 = (x_1 + x_2)^2 + x_2^2,$$

  which is 0 only if $x_2 = 0$ and $x_1 = 0$.
- A non-example: $\langle x, y \rangle = x_1 y_2 + x_2 y_1$ on $\mathbb{R}^2$ is linear and symmetric, but $\langle (1, -1), (1, -1) \rangle = -2 < 0$.

§2.5 The norm and the Cauchy–Schwarz inequality (Ch. 17)

Definition (Def. 44)**.** A **norm** on V is a function $\| \cdot \| \colon V \to \mathbb{R}$ with (i) $\|v\| > 0$ for $v \neq 0$; (ii) $\|\lambda v\| = |\lambda|\, \|v\|$; (iii) the **triangle inequality** $\|u + v\| \leq \|u\| + \|v\|$.

An inner product gives the candidate $\|v\| = \sqrt{\langle v, v \rangle}$, the length. (For the dot product on $\mathbb{R}^2$ this is $\sqrt{x_1^2 + x_2^2}$.) Two facts about it are needed first. Vectors u, v are **orthogonal** if $\langle u, v \rangle = 0$.

Pythagoras. If $\langle u, v\rangle = 0$, then $\|u+v\|^2 = \|u\|^2 + \|v\|^2$. Proof: expand by linearity in each slot,

$$\|u+v\|^2 = \langle u+v, u+v\rangle = \langle u,u\rangle + \langle u,v\rangle + \langle v,u\rangle + \langle v,v\rangle = \|u\|^2 + 0 + 0 + \|v\|^2.$$

Splitting off the part along v. If $v \neq 0$ and $c = \langle u, v\rangle / \|v\|^2$, then $w = u - cv$ is orthogonal to v. Proof: $\langle w, v\rangle = \langle u, v\rangle - c\langle v, v\rangle = \langle u, v\rangle - \langle u, v\rangle = 0$. So $u = cv + w$ is u split into a piece along v and a piece perpendicular to v.

Theorem (Prop. 82 / LADR 6.14: Cauchy–Schwarz inequality)
For all $u, v \in V$, $|\langle u, v\rangle| \leq \|u\|\,\|v\|$, with equality if and only if one of u, v is a scalar multiple of the other.

The idea

In the plane, $|\langle u, v\rangle| = \|u\|\,\|v\|\,|\cos\theta|$, and $|\cos\theta| \leq 1$: the shadow of u on the line through v is no longer than u. To prove this without angles, split u into its piece cv along v and its perpendicular piece w. By Pythagoras $\|u\|^2 = \|cv\|^2 + \|w\|^2 \geq \|cv\|^2$, and $\|cv\|$ turns out to be exactly $|\langle u, v\rangle|/\|v\|$. Equality means the perpendicular piece is 0.

Proof. 1. If $v = 0$, both sides are 0, and $v = 0 \cdot u$ is a multiple of u. From now on $v \neq 0$.

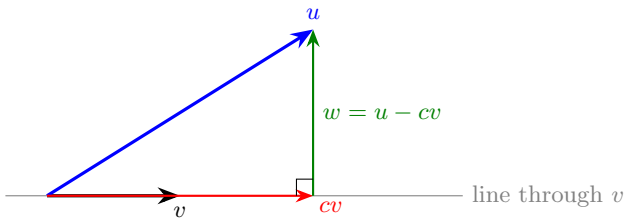
2. Let $c = \langle u, v\rangle / \|v\|^2$ and $w = u - cv$. Then $\langle w, v\rangle = 0$, so cv and w are orthogonal, and Pythagoras gives

$$\|u\|^2 = \|cv\|^2 + \|w\|^2 = |c|^2\|v\|^2 + \|w\|^2 \geq |c|^2\|v\|^2 = \frac{|\langle u, v\rangle|^2}{\|v\|^2}.$$

3. Multiply by $\|v\|^2$ and take square roots: $|\langle u, v\rangle| \leq \|u\|\,\|v\|$.

4. Equality holds exactly when $\|w\|^2 = 0$, that is, when $u = cv$. Conversely, if $u = av$ for a scalar a, then $|\langle u, v\rangle| = |a|\,\|v\|^2 = \|u\|\,\|v\|$.

(The notes expand $\langle z, z\rangle \geq 0$ for $z = cv - u = -w$; it is the same computation.) □



$u = cv + w$ with $w \perp v$. The hypotenuse u is at least as long as the side cv, and $\|cv\| = |c|\,\|v\| = |\langle u, v\rangle|/\|v\|$. That is the Cauchy–Schwarz inequality. Equality means $w = 0$: u lies on the line through v.

Ex. 67: bounding an integral. The function $\sqrt{\sin x}$ has no elementary antiderivative; numerically its integral over $[0, \pi]$ is about 2.396. With the L^2 product on $[0, \pi]$, $f = \sqrt{\sin x}$ and $g = 1$:

$$\int_0^\pi \sqrt{\sin x}\,dx = \langle f, g\rangle \leq \|f\|\,\|g\| = \sqrt{\int_0^\pi \sin x\,dx} \cdot \sqrt{\int_0^\pi 1\,dx} = \sqrt{2} \cdot \sqrt{\pi} = \sqrt{2\pi} \approx 2.507.$$

Theorem (Prop. 81 / LADR 6.7, 6.17: an inner product gives a norm)
$\|v\| = \sqrt{\langle v, v\rangle}$ is a norm on V.

The idea

Positivity and scaling come straight from the axioms. For the triangle inequality, expand $\|u+v\|^2$: the cross terms add up to $2\,\mathrm{Re}\langle u, v\rangle$, and Cauchy–Schwarz bounds them by $2\|u\|\,\|v\|$, which is exactly the cross term of $(\|u\| + \|v\|)^2$.

Proof. 1. *Positive.* If $v \neq 0$ then $\langle v, v\rangle > 0$, so $\|v\| > 0$.

2. *Scaling.* Using linearity in the first slot and conjugate-linearity in the second,

$$\|\lambda v\|^2 = \langle \lambda v, \lambda v\rangle = \lambda\bar{\lambda}\langle v, v\rangle = |\lambda|^2\|v\|^2.$$

   Take square roots.

3. *Triangle inequality.* Expand, using $\langle v, u\rangle = \overline{\langle u, v\rangle}$ and $z + \bar{z} = 2\operatorname{Re} z$ (T4):

$$\|u+v\|^2 = \|u\|^2 + \langle u, v\rangle + \langle v, u\rangle + \|v\|^2 = \|u\|^2 + 2\operatorname{Re}\langle u, v\rangle + \|v\|^2.$$

4. Since $\operatorname{Re} z \le |z|$ (T4) and by Cauchy–Schwarz,

$$\|u\|^2 + 2\operatorname{Re}\langle u, v\rangle + \|v\|^2 \le \|u\|^2 + 2\|u\|\,\|v\| + \|v\|^2 = \big(\|u\| + \|v\|\big)^2.$$

   Take square roots.

□

Equality in the triangle inequality needs equality in both steps of 4: u, v dependent and $\operatorname{Re}\langle u, v\rangle = |\langle u, v\rangle|$. For $v \neq 0$ that means $u = av$ with $a \ge 0$: the two vectors point the same way.

§2.6 Which norms come from inner products (Ch. 17)

Theorem (Prop. 83 / LADR 6.21, 6A.28: parallelogram law and polarization)
A norm $\|\cdot\|$ on V comes from an inner product if and only if it satisfies the **parallelogram law**

$$\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2 \quad \text{for all } u, v.$$

In that case the inner product is recovered from the norm by the **polarization identity**

$$\langle u, v\rangle = \frac{\|u+v\|^2 - \|u-v\|^2}{4} \quad (\mathbb{F} = \mathbb{R}), \qquad \langle u, v\rangle = \frac{\|u+v\|^2 - \|u-v\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2}{4} \quad (\mathbb{F} = \mathbb{C}).$$

The idea

Geometrically: the two diagonals of a parallelogram have squared lengths adding up to the squared lengths of the four sides. For a norm from an inner product, expanding $\|u \pm v\|^2$ gives the same cross term with opposite signs; adding cancels it (parallelogram law) and subtracting isolates it (polarization). The converse, that the law is enough to define an inner product, is the Jordan–von Neumann theorem; the notes do not prove it.

Proof of the parallelogram law and polarization for a norm from an inner product. 1. As in the proof of Prop. 81,

$$\|u+v\|^2 = \|u\|^2 + 2\operatorname{Re}\langle u, v\rangle + \|v\|^2, \qquad \|u-v\|^2 = \|u\|^2 - 2\operatorname{Re}\langle u, v\rangle + \|v\|^2.$$

2. Adding the two lines gives the parallelogram law.

3. Subtracting gives $\|u+v\|^2 - \|u-v\|^2 = 4\operatorname{Re}\langle u, v\rangle$. Over $\mathbb{R}$, $\langle u, v\rangle$ is real, which is the real polarization identity.

4. Over $\mathbb{C}$, apply step 3 to u and iv, using $\langle u, iv\rangle = \bar{i}\langle u, v\rangle = -i\langle u, v\rangle$ and $\operatorname{Re}(-iz) = \operatorname{Im} z$:

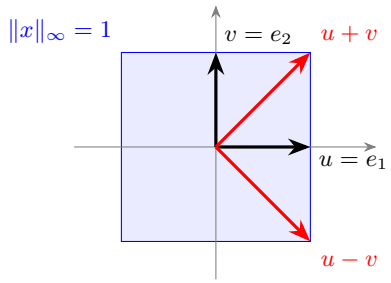
$$\|u+iv\|^2 - \|u-iv\|^2 = 4\operatorname{Re}\langle u, iv\rangle = 4\operatorname{Re}\big(-i\langle u, v\rangle\big) = 4\operatorname{Im}\langle u, v\rangle.$$

   So the numerator of the complex formula is $4\operatorname{Re}\langle u, v\rangle + i \cdot 4\operatorname{Im}\langle u, v\rangle = 4\langle u, v\rangle$.

□

Check of the real formula with $u = (1, 2)$, $v = (3, -1)$: $\|u+v\|^2 = \|(4, 1)\|^2 = 17$, $\|u-v\|^2 = \|(-2, 3)\|^2 = 13$, and $(17 - 13)/4 = 1 = u \cdot v$.

A norm that does not come from an inner product. The **maximum norm** $\|(z_1,\dots,z_n)\|_\infty = \max_j |z_j|$ satisfies (i)–(iii) of Def. 44. For $u = e_1$, $v = e_2$ in $\mathbb{R}^2$: $\|u \pm v\|_\infty = \|(1, \pm 1)\|_\infty = 1$, so the left side of the parallelogram law is 2, while the right side is $2 + 2 = 4$.



The unit sphere of the maximum norm on $\mathbb{R}^2$ is the square. All four vectors $u, v, u+v, u-v$ lie on it, so all have norm 1: the diagonals are as long as the sides, and the parallelogram law fails ($1 + 1 \neq 2 + 2$).

§2.7 Angles (Ch. 17)

Definition (Def. 45 / LADR 6A.16)**.** In a *real* inner product space the **angle** $\theta \in [0, \pi]$ between nonzero u, v is defined by $\cos\theta = \langle u, v\rangle / (\|u\|\,\|v\|)$.

Cauchy–Schwarz is exactly the statement that the right side lies in $[-1, 1]$, so θ exists. Ex. 68: for $(1, 0)$ and $(1, 1)$ in $\mathbb{R}^2$, $\cos\theta = 1/(1 \cdot \sqrt{2})$, so $\theta = \pi/4$. In a complex space $\langle u, v\rangle$ need not be real, so this formula does not define an angle without an extra convention.

§2.8 Optional application: attention in language models (Ch. 17)

This section is not needed later in the course. It shows inner products, linear maps and stochastic matrices working together inside a transformer.

The problem: retrieval by similarity. A memory holds m items. Item j has a **key** $k_j \in \mathbb{R}^{d_k}$, used to decide whether the item is relevant, and a **value** $\nu_j \in \mathbb{R}^{d_v}$, the information it supplies if it is used. The key and the value have different jobs, so they may have different dimensions. A request is a **query** $q \in \mathbb{R}^{d_k}$, in the same space as the keys because it is compared with them. The comparison is the inner product: a large $\langle q, k_j\rangle$ means item j matches.

What the score measures. For real vectors $\langle q, k\rangle = \|q\|\,\|k\|\cos\theta$ (Def. 45). So the score is large when k points in nearly the same direction as q, and it also grows with the lengths of q and k. It is not the angle alone; normalizing both vectors to length 1 would turn it into $\cos\theta$, but attention uses the inner product itself.

From scores to weights: softmax. We want to combine the values with weights that are positive, add up to 1, and are larger for larger scores. With scores $s_1, \dots, s_m$, the **softmax** weights are

$$a_j = \frac{e^{s_j}}{e^{s_1} + \cdots + e^{s_m}}.$$

- Each $a_j > 0$, because $e^s > 0$; and $a_1 + \cdots + a_m = 1$, because the numerators add up to the denominator.
- The ratio of two weights depends only on the difference of scores: $a_j / a_l = e^{s_j - s_l}$. So larger scores get larger weights, and adding the same constant to every score changes nothing.
- Multiplying all scores by a large factor makes the largest score dominate: with scores 10 and 0, the weights are $e^{10}/(e^{10}+1) \approx 0.99995$ and 0.00005. So softmax is a smooth version of "pick the item with the largest score". That is the reason for the name.

The output is the weighted average of the values, $y = a_1\nu_1 + \cdots + a_m\nu_m$. Several items can contribute, in proportion to how well they match. Scaled dot-product attention uses the scores $s_j = \langle q, k_j\rangle / \sqrt{d_k}$; the factor $\sqrt{d_k}$ is explained below.

Ex. 69: a toy memory, computed in full. Three records France $\mapsto$ Paris, Germany $\mapsto$ Berlin, Italy $\mapsto$ Rome. Choose the values $\nu_1 = e_{\text{Paris}}$, $\nu_2 = e_{\text{Berlin}}$, $\nu_3 = e_{\text{Rome}}$ (the standard basis of $\mathbb{R}^3$), the keys $k_1 = (2, 0)$, $k_2 = (0, 1)$,

$k_3 = (-1, 0)$ in $\mathbb{R}^2$, and the query $q = (1, 0)$ for France. These coordinates are chosen by hand so that every step can be seen; they are not how a real model encodes countries.

1. Inner products: $\langle q, k_1\rangle = 2$, $\langle q, k_2\rangle = 0$, $\langle q, k_3\rangle = -1$.
2. Here $d_k = 2$, so the scores are $2/\sqrt{2} = \sqrt{2}$, 0, $-1/\sqrt{2}$.
3. Exponentials: $e^{\sqrt{2}} \approx 4.113$, $e^0 = 1$, $e^{-1/\sqrt{2}} \approx 0.493$, with sum ≈ 5.606. Dividing gives the weights 0.734, 0.178, 0.088.
4. Output: $y \approx 0.734\, e_{\text{Paris}} + 0.178\, e_{\text{Berlin}} + 0.088\, e_{\text{Rome}}$. The best match dominates, but the others still contribute.



Left: the query and the three keys. k_1 points the same way as q and is long, so its score is largest; k_2 is perpendicular to q (score 0); k_3 points the opposite way (negative score). Right: after softmax every weight is positive and they add up to 1.

Why divide by $\sqrt{d_k}$. As the dimension d_k grows, inner products get larger, and softmax of large scores is nearly "all weight on one item". A heuristic shows how fast they grow. Suppose the coordinates q_r and k_r behave like independent random numbers with mean 0 and variance 1.

1. Each product $q_r k_r$ has mean $E[q_r]E[k_r] = 0$ and variance $E[q_r^2 k_r^2] = E[q_r^2]E[k_r^2] = 1$, using independence.
2. The d_k products are independent, and variances of independent terms add, so $\langle q, k\rangle = \sum_{r=1}^{d_k} q_r k_r$ has variance d_k: a typical score has size about $\sqrt{d_k}$.
3. Dividing by $\sqrt{d_k}$ gives variance 1, so typical scores stay of size about 1 in every dimension, and softmax does not collapse onto a single item.

Learned queries, keys and values. In a transformer, position j of the text has a current internal representation $x_j \in \mathbb{R}^{d_{\text{model}}}$. One attention head learns three linear maps $W_Q, W_K \colon \mathbb{R}^{d_{\text{model}}} \to \mathbb{R}^{d_k}$ and $W_V \colon \mathbb{R}^{d_{\text{model}}} \to \mathbb{R}^{d_v}$, and sets $q_i = W_Q x_i$, $k_j = W_K x_j$, $\nu_j = W_V x_j$. So each position offers a key (how it should be matched) and a value (what it contributes), and asks with a query. The keys decide where information flows from; the values are the information that flows.

Self-attention. Position i compares its query with every key:

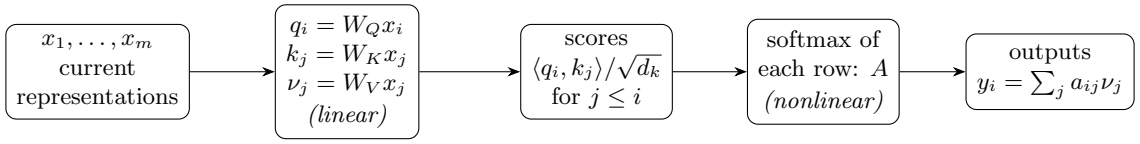
$$a_{ij} = \frac{\exp\big(\langle q_i, k_j\rangle/\sqrt{d_k}\big)}{\sum_{\ell=1}^{m} \exp\big(\langle q_i, k_\ell\rangle/\sqrt{d_k}\big)}, \qquad y_i = \sum_{j=1}^{m} a_{ij}\nu_j.$$

- For each i, $a_{ij} > 0$ and $\sum_j a_{ij} = 1$, so the matrix $A = (a_{ij})$ is *row*-stochastic: row i says how position i spreads its attention. (The Markov chains of Weeks 6–7 used column-stochastic matrices.)
- *Masking.* A model that writes text left to right must not let position i use later positions. So the terms with $j > i$ are removed before softmax: row i has nonzero entries only in columns $1, \dots, i$.
- *Matrix form.* Put the vectors as columns: $Q = (q_1 \cdots q_m)$, $K = (k_1 \cdots k_m)$, $V_{\text{val}} = (\nu_1 \cdots \nu_m)$. Then $S = Q^{\mathsf{T}} K/\sqrt{d_k}$ has entries $S_{ij} = \langle q_i, k_j\rangle/\sqrt{d_k}$, and A is softmax applied to each row of S. Column i of $Y = V_{\text{val}} A^{\mathsf{T}}$ is V_{val} times column i of A^{T}, which is row i of A; so it is $\sum_j a_{ij}\nu_j = y_i$. Once the weights are known, the retrieval is ordinary matrix multiplication.
- *Not linear.* A is computed from the inputs, and softmax is not linear. For instance, doubling every x_j doubles every q_i and k_j, so every score is multiplied by 4 and the weights change. Self-attention as a whole is therefore not a linear map, although each step uses linear maps.

A two-position example with masking. Take $W_Q = W_K = W_V = I$ on $\mathbb{R}^2$, $x_1 = (1,0)$, $x_2 = (1,1)$, so $d_k = 2$. Row 1 may use only position 1, so $a_{11} = 1$ and $y_1 = x_1$. Row 2 has scores $\langle x_2, x_1\rangle/\sqrt{2} = 1/\sqrt{2}$ and $\langle x_2, x_2\rangle/\sqrt{2} = 2/\sqrt{2}$, with weights

$$a_{21} = \frac{e^{1/\sqrt{2}}}{e^{1/\sqrt{2}} + e^{2/\sqrt{2}}} \approx 0.330, \qquad a_{22} \approx 0.670, \qquad y_2 \approx 0.330\, x_1 + 0.670\, x_2 = (1,\, 0.670).$$

If x_1, x_2 are doubled, the row-2 scores become $4/\sqrt{2}$ and $8/\sqrt{2}$, the weights become 0.056 and 0.944, and $y_2 \approx (2, 1.89)$, which is not twice $(1, 0.670)$.



One attention head. Learned linear maps produce queries, keys and values; inner products of queries with keys decide the weights; the weights mix the values. Only the softmax step is nonlinear.

Shortcuts

- **Trace from eigenvalues** (over $\mathbb{C}$): $\operatorname{tr} T = \sum$ eigenvalues with algebraic multiplicity; for a nilpotent T this is 0. For a real matrix, a non-real eigenvalue λ comes with $\bar{\lambda}$, contributing $2 \operatorname{Re} \lambda$.
- **Trace from χ_T**: $-\operatorname{tr} T$ is the coefficient of z^{n-1}.
- **Trace of a projection** ($P^2 = P$): $\operatorname{tr} P = \dim \operatorname{range} P$, by choosing a basis of $\operatorname{null} P$ followed by a basis of $\operatorname{range} P$ (HW9 P1).
- **Impossible commutator**: to show $ST - TS = X$ has no solution, check whether $\operatorname{tr} X \neq 0$.
- **Cauchy–Schwarz with a chosen vector**: for real numbers, $(a_1 b_1 + \cdots + a_n b_n)^2 \le (\sum a_i^2)(\sum b_i^2)$. With $b = (1, \ldots, 1)$: $(\sum a_i)^2 \le n \sum a_i^2$. With weights: $(a + 2b + 3c)^2 \le (1 + 4 + 9)(a^2 + b^2 + c^2)$. For integrals: $\left(\int fg\right)^2 \le \int f^2 \int g^2$, often with $g = 1$.
- **Equality in Cauchy–Schwarz** $\iff$ the two vectors are dependent: say which multiple.
- **Is it an inner product?** Test positivity first, on a vector that makes a cross term negative, such as $(1, -1)$. Completing the square proves positivity.
- **Recover an inner product from a norm** by polarization; test whether a norm can come from one by the parallelogram law with $u = e_1$, $v = e_2$.
- **Softmax**: shifting all scores changes nothing; only differences matter, $a_j/a_l = e^{s_j - s_l}$.

Traps

- $\operatorname{tr}(AB) \neq \operatorname{tr} A \cdot \operatorname{tr} B$ in general, and $\operatorname{tr}(ABC)$ may differ from $\operatorname{tr}(ACB)$: only cyclic moves are allowed.
- Over $\mathbb{C}$ the second slot is conjugate-linear: $\langle u, \lambda v\rangle = \bar{\lambda}\langle u, v\rangle$, not $\lambda\langle u, v\rangle$.
- Linear and symmetric is not enough for an inner product; positivity must be checked on every nonzero vector.
- $\|u + v\|^2 = \|u\|^2 + \|v\|^2$ needs $\langle u, v\rangle = 0$ (over $\mathbb{C}$, $\operatorname{Re}\langle u, v\rangle = 0$ suffices for this identity, but orthogonality means $\langle u, v\rangle = 0$).
- The triangle inequality is an equality only when one vector is a *nonnegative* multiple of the other.
- $ST - TS = I$ is impossible only in finite dimensions ($DM - MD = I$ on $\mathcal{P}(\mathbb{R})$).
- The attention weights are not a linear function of the inputs.

§3 Problem set

HW9 problems as assigned; then problems I1–I7 on the same material.

HW9 P1. Let P be a projection on a finite-dimensional vector space, so that $P^2 = P$. Prove that $\operatorname{tr} P = \operatorname{rank} P$. You may choose a basis adapted to the decomposition $V = \operatorname{null} P \oplus \operatorname{range} P$.

HW9 P2. Let $T \in \mathcal{L}(\mathbb{C}^5)$ have minimal polynomial $m_T(x) = (x-2)^2(x+1)$.

(a) Determine all possible algebraic multiplicities of the eigenvalues 2 and -1.

(b) Determine all possible values of $\operatorname{tr} T$.

(c) For each possible trace, exhibit a Jordan normal form realizing it.

HW9 P3.

(a) Prove that every nilpotent operator on a finite-dimensional complex vector space has trace 0.

(b) Give a nonzero operator with trace 0 that is not nilpotent.

(c) Give two 2×2 matrices A, B for which $AB - BA \neq 0$, and verify directly that $\operatorname{tr}(AB - BA) = 0$.

HW9 P4. For each of the following formulas on $\mathbb{R}^2$, decide whether it defines an inner product. Justify your answer.

$$\langle x, y\rangle_1 = x_1y_1 + 2x_2y_2, \qquad \langle x, y\rangle_2 = x_1y_1 - x_2y_2, \qquad \langle x, y\rangle_3 = (x_1 + x_2)(y_1 + y_2).$$

HW9 P5. On $\mathcal{P}_2(\mathbb{R})$ define $\langle p, q\rangle = \int_{-1}^{1} p(x)q(x)\,dx$.

(a) Prove that this is an inner product.

(b) Compute the norms of 1, x, and x^2.

(c) Determine which pairs among $1, x, x^2$ are orthogonal.

HW9 P6. Use the Cauchy–Schwarz inequality to prove that for all real numbers a, b, c, $(a+b+c)^2 \le 3(a^2+b^2+c^2)$. Determine exactly when equality holds.

HW9 P7. (optional) Prove the parallelogram identity $\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$ in every real or complex inner-product space.

I1 (the trace inner product). For real $n \times n$ matrices define $\langle A, B\rangle = \operatorname{tr}(A^{\mathsf{T}}B)$.

(a) Show that $\langle A, A\rangle = \sum_{i,j} a_{ij}^2$, and deduce that $A^{\mathsf{T}}A = 0$ only if $A = 0$.

(b) Show that $\langle \cdot, \cdot \rangle$ is an inner product on $\mathbb{R}^{n\times n}$.

(c) Use Cauchy–Schwarz to show $\big(\operatorname{tr}(A^{\mathsf{T}}B)\big)^2 \le \operatorname{tr}(A^{\mathsf{T}}A)\operatorname{tr}(B^{\mathsf{T}}B)$.

I2 (trace of a real matrix). A real 3×3 matrix A has eigenvalues 2 and $1+i$. Find $\operatorname{tr} A$. If moreover A is invertible, can $\operatorname{tr} A^{-1}$ be found from this information?

I3 (commutators). (a) Find 2×2 matrices A, B with $AB - BA = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. (b) Show there are no 2×2 matrices with $AB - BA = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$.

I4 (Cauchy–Schwarz). (a) For real $x_1, \dots, x_n$, prove that the square of the average is at most the average of the squares. (b) For continuous $f\colon [0,1] \to \mathbb{R}$, prove

$$\Big(\int_0^1 x f(x)\,dx\Big)^2 \le \frac{1}{3}\int_0^1 f(x)^2\,dx,$$

and find all f for which equality holds.

I5 (polarization). The function $\|(x, y)\| = \sqrt{x^2 + xy + y^2}$ is a norm on $\mathbb{R}^2$ that satisfies the parallelogram law. Use polarization to find the inner product it comes from, and check that the result is an inner product.

I6 (angles in high dimensions). In $\mathbb{R}^n$ with the dot product, find the angle between $(1, 1, \ldots, 1)$ and e_1. What happens as $n \to \infty$?

I7 (attention). In the toy memory of Ex. 69, (a) compute the weights for the query $q = (0, 1)$; (b) show that replacing q by tq with $t \to \infty$ makes the weights for the query $(1, 0)$ tend to $(1, 0, 0)$; (c) show that adding the same vector w to all three keys does not change the weights, but multiplying all three keys by 2 does.